

# Global Valuation

## Cross-Country Valuation Multiples: Data and First Results

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Augustin Landier

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HEC Paris

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# Data

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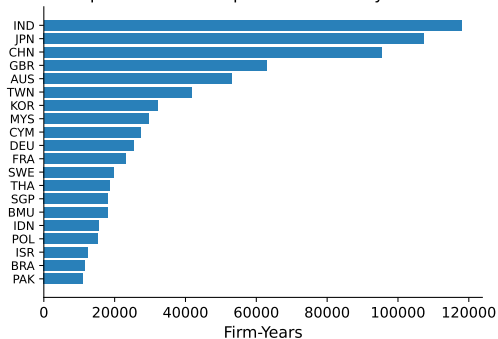
# Datasets

| File              | Size   | Rows  | Cols | Description                                |
|-------------------|--------|-------|------|--------------------------------------------|
| compustat_global  | 2.3 GB | 981K  | 468  | Global annual fundamentals (124 countries) |
| compustat_america | 0.8 GB | 254K  | 984  | NA fundamentals + CRSP link                |
| returns_global    | 1.9 GB | 10.4M | 25   | Global monthly prices                      |
| returns_america   | 0.9 GB | 3.6M  | 55   | CRSP US monthly returns                    |

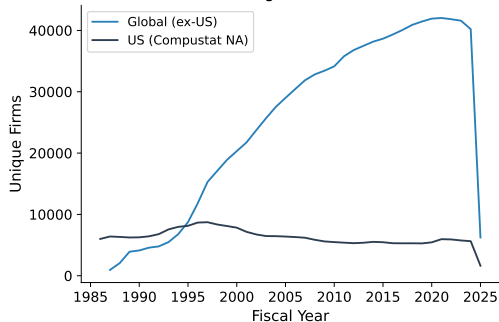
- Compustat Global: fiscal years 1987–2025, 124 countries
- Compustat America: fiscal years 1986–2025, CRSP-linked
- Combined regression panel: **749,527 firm-years**, 72,914 firms, 81 countries

# Coverage: Countries and Time

Compustat Global: Top 20 Countries by Firm-Years



Coverage Over Time



India, Japan, China dominate non-US coverage. Panel peaks at ~41K firms (2022).

## Panel Construction

- **Market-to-book:**  $\log(M/B) = \log(\text{Market Cap}/\text{Book Equity})$ 
  - US: `mkvalt` from Compustat NA; Global: `cshoi`  $\times$  `prccm`
- **ROA:** Net Income / Total Assets    **Size:**  $\log(\text{Total Assets})$
- **Lag earnings growth:**  $(E_{t-1} - E_{t-2})/|E_{t-2}|$     **Industry:** SIC-2
- Winsorize M/B at 1st/99th percentiles; filter country-years with  $\geq 30$  obs

| Variable              | N       | Mean  | P10   | Median | P90   |
|-----------------------|---------|-------|-------|--------|-------|
| $\log(M/B)$           | 749,527 | 0.46  | -0.77 | 0.39   | 1.76  |
| ROA                   | 749,093 | -0.02 | -0.15 | 0.02   | 0.11  |
| $\log(\text{Assets})$ | 749,527 | 7.28  | 3.26  | 7.17   | 11.49 |

# Cross-Country Regressions

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# Regression Specifications

## Market-to-Book:

$$\begin{aligned}\log(M/B)_{i,t} = & \alpha + \beta_1 \text{ROA}_{i,t} + \beta_2 \log(\text{Assets})_{i,t} + \beta_3 \Delta E_{i,t-1} \\ & + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}\end{aligned}\quad (\text{Spec 1})$$

$$\begin{aligned}\log(M/B)_{i,t} = & \alpha + \beta_1 \text{ROA}_{i,t} + \beta_2 \log(\text{Assets})_{i,t} + \beta_3 \Delta E_{i,t-1} \\ & + \delta_j + \gamma_{c,t} + \varepsilon_{i,t}\end{aligned}\quad (\text{Spec 2})$$

## ROA:

$$\text{ROA}_{i,t} = \alpha + \beta_1 \log(\text{Assets})_{i,t} + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}$$

$\gamma_c$  = country FE,  $\delta_j$  = industry FE (SIC-2),  $\tau_t$  = year FE,  $\gamma_{c,t}$  = country  $\times$  year FE.  
Standard errors clustered at the country level.



## M/B Regression Results

|                   | <b>Spec 1</b>       | <b>Spec 2</b>       |
|-------------------|---------------------|---------------------|
|                   | C + Ind + Year FE   | C×Year + Ind FE     |
| ROA               | −0.070<br>(0.133)   | −0.102<br>(0.129)   |
| log(Assets)       | −0.030<br>(0.024)   | −0.032<br>(0.024)   |
| Lag Earn. Growth  | 0.006***<br>(0.002) | 0.005***<br>(0.002) |
| N / $R^2$         | 607,209 / 0.185     | 607,209 / 0.211     |
| C / Ind / Year FE | Y / Y / Y           | − / Y / −           |
| C × Year FE       | −                   | Yes                 |

\*\*\*  $p < 0.01$ . Clustered SEs at country level.

## Key Takeaway: ROA Does Not Explain M/B

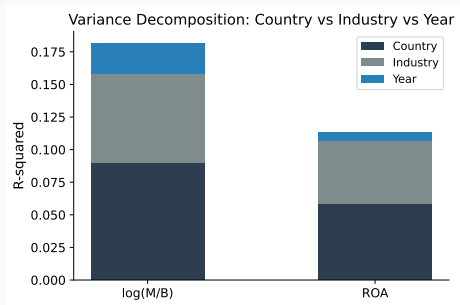
- ROA coefficient is **insignificant** in both specifications
- After country, industry, and year controls, current profitability does not predict M/B
- Only **lag earnings growth** is significant (positive, as expected)
- Size effect negative but insignificant with clustered SEs
- **Interpretation:** cross-country M/B differences reflect *discount rates* and *growth expectations*, not current profitability

$R^2$  **comparison:** Spec 1 = 0.185, Spec 2 = 0.211 — country×year interactions gain 2.6 pp, suggesting time-varying country effects matter.

# Variance Decomposition

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# What Drives Valuation? Country vs Industry vs Year



## Shapley $R^2$ decomposition:

### log(M/B):

- Country: **49.6%** of  $R^2$
- Industry: 37.7%
- Year: 12.7%

### ROA:

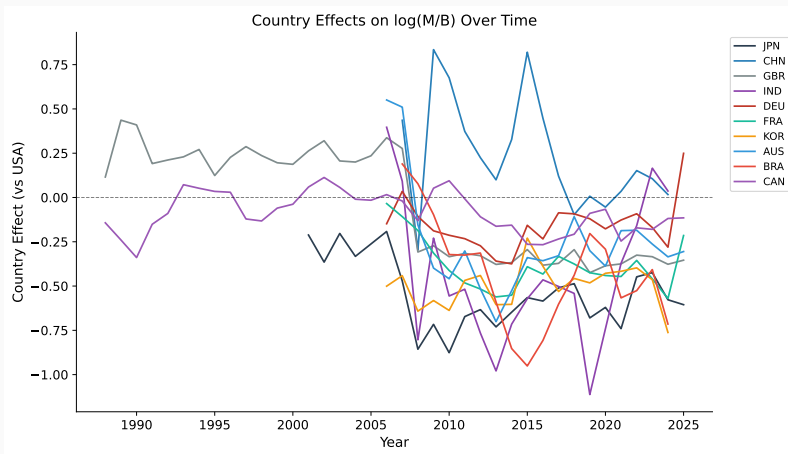
- Country: **51.2%** of  $R^2$
- Industry: 43.2%
- Year: 5.6%

Country dominates both valuation and profitability.

## Country Effects Over Time

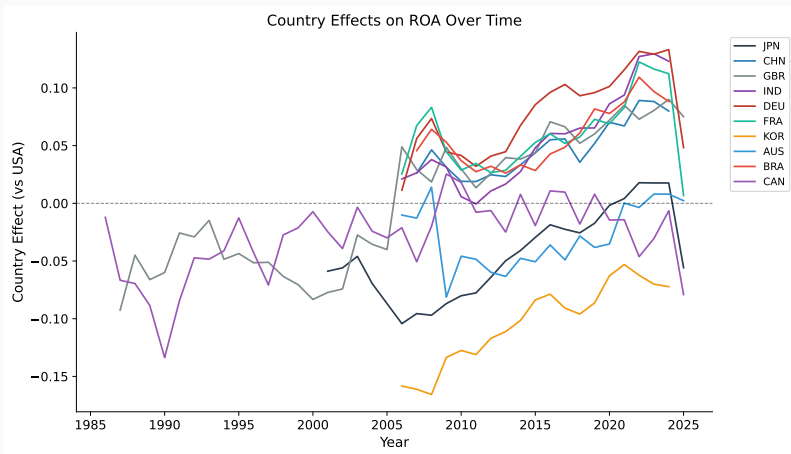
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# Country Effects on M/B (vs USA)



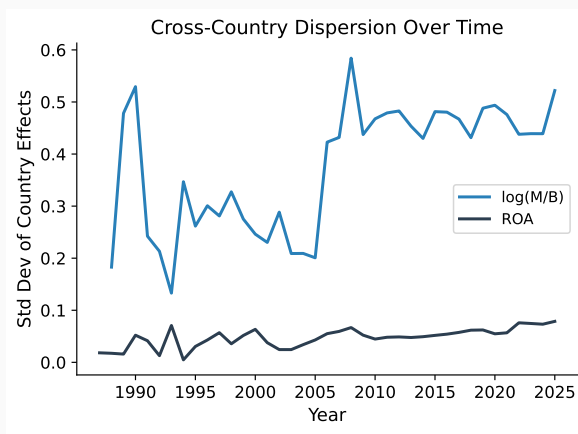
Year-by-year:  $\log(M/B) \sim \text{ROA} + \log(\text{Assets}) + \text{Country FE} + \text{Industry FE}$ . USA = 0.

# Country Effects on ROA (vs USA)



Year-by-year:  $ROA \sim \log(\text{Assets}) + \text{Country FE} + \text{Industry FE}$ . USA = 0.

# Cross-Country Dispersion Over Time



## No convergence in valuations:

- M/B dispersion:  $0.40 \rightarrow 0.47$
- Trend is **diverging**
- Spike during 2008–09 GFC

## ROA dispersion more stable:

- Lower level ( $\sim 0.05$ – $0.10$ )
- Spike in 2009–10 (earnings collapse)

Valuation gaps are *widening* despite globalization.



## Stylized Facts

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## Stylized Facts (1/2): Decomposition

1. **Country effects dominate valuation** — country explains 50% of M/B  $R^2$  (Shapley), vs 38% for industry. Echoes the Heston–Rouwenhorst debate, but for *levels* rather than returns.
2. **Country dominates profitability too** — for ROA, country edges industry (51% vs 43%). Both valuation and profitability are primarily location-driven.
3. **Negative M/B–ROA correlation across countries** ( $\rho = -0.26$ ) — high-M/B countries are *not* high-ROA countries. Discount rates and growth expectations, not current profitability, drive the cross-country valuation map.

## Stylized Facts (2/2): Dynamics

4. **Japan discount is persistent** — M/B effect negative in **100%** of years (avg  $-0.54$  vs USA), with no sign of convergence.
5. **China premium collapsed post-2015** — from  $+0.8$  to near zero, consistent with market correction and deleveraging.
6. **Cross-country valuation dispersion is increasing** — despite financial globalization, countries are *diverging* in M/B (std dev:  $0.40 \rightarrow 0.47$ ).

# Literature

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## Related Literature: Decomposition and Institutions

### Country vs industry effects:

- Heston–Rouwenhorst (1994, *JFE*) — country effects dominate for returns. We find the same for *valuation levels*.
- Griffin–Karolyi (1998, *JFE*) — industry effects matter more in traded-goods sectors.
- Brooks–Del Negro (2004) — country effects declined post-1999 for returns. Opposite for M/B.

### Legal origin and investor protection:

- La Porta et al. (2002, *JF*) — investor protection raises Tobin's Q.
- Djankov et al. (2008, *JFE*) — anti-self-dealing index predicts valuations.
- Doidge et al. (2007, *JFE*) — countries matter more than firms for governance.

## Related Literature: Globalization and Dynamics

### Limits of convergence:

- Stulz (2005, *JF*) — “twin agency problems” prevent equalization.
- Bekaert et al. (2011, *RFS*) — segmentation decreased but persists.
- Doidge et al. (2004, *JFE*) — cross-listed firms gain  $\sim 17\%$  Q premium.

### Country-specific puzzles and fundamentals:

- French–Poterba (1991, *JFE*) — Japan P/E premium; persistent discount since.
- Fama–French (1998, *JF*) — value premium exists internationally.
- Rajan–Zingales (1998, *AER*) — financial development raises valuations.

## Our Contribution

- First panel decomposition of *valuation levels* (not returns) into country, industry, and time using Shapley  $R^2$
- Largest cross-country valuation panel: 81 countries, 1986–2025, 749K firm-years
- Document **divergence** in M/B across countries, challenging the globalization-implies-equalization view
- Country M/B effects **not** explained by ROA ( $\rho = -0.26$ ) — points to discount-rate and institutional channels